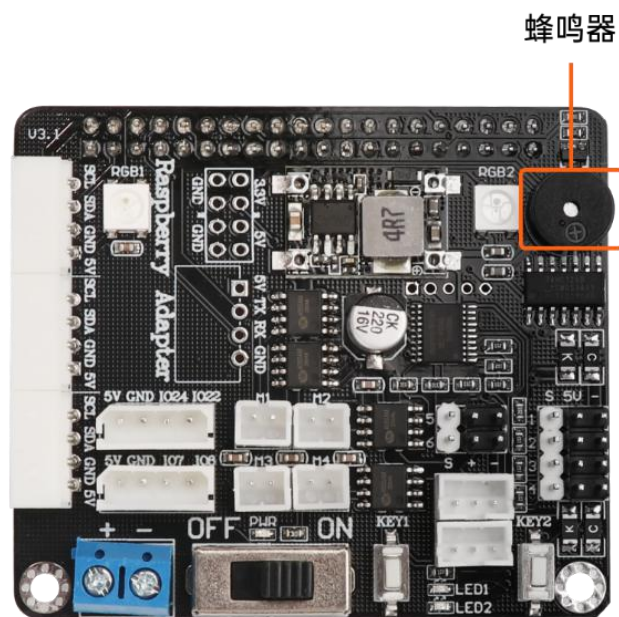


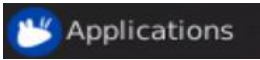
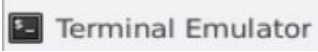
Lesson 4 Buzzer Control

1. Preparation

There is a buzzer on the Raspberry Pi expansion board, as shown in the figure below:



2. Operation Step

- 1) Open LX terminal. Click  and select  to open command line terminal of Ubuntu.
- 2) Initially, enter “**rostopic pub /sensor/buzzer std_msgs/Float32 "data: 0.0"**” command and complete the command through TAB key.

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ rostopic pub /sensor/buzzer std_msgs/Float32 "data: 0.0"
```

- 3) Next, revise “data:0.0” in the command. For example, we change 0.00 to 0.5. And press Enter.

A screenshot of a terminal window titled "Terminal - ubuntu@ubuntu: ~". The terminal shows the command `rostopic pub /sensor/buzzer std_msgs/Float32 "data: 0.5"` being entered at the prompt `ubuntu@ubuntu:~$`. The terminal has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help".

- 4) If you want to quit the program, please press “Ctrl+C”.

3. Project Outcome

After the program starts, the buzzer on the expansion board will first beep lasting for 0.5 seconds, and then mute automatically.